

The Director General of Agriculture,

Department of Agriculture,

Peradeniya,

Sri Lanka.

REQUEST FOR SOIL TYPE DATASET FOR A FINAL-YEAR UNDERGRADUATE RESEARCH PROJECT – “INTEGRATED FLOOD AND LANDSLIDE RISK ASSESSMENT AND RESPONSE SYSTEM FOR SRI LANKA” (PROJECT ID: J26-IT-334)

Dear Sir/Madam,

We are final-year BSc (Hons) in Information Technology undergraduates at the Sri Lanka Institute of Information Technology (SLIIT), specialising in Artificial Intelligence under the Center of Excellence for AI (CEAI). We are jointly conducting our final-year research project titled “**Integrated Flood and Landslide Risk Assessment AND Response System for Sri Lanka**” (Project ID: J26-IT-334), under the supervision of Dr. Kapila Dissanayaka (Supervisor) and Mr. Uchira Wickramaratne (Co-Supervisor).

The project develops an AI-driven, multi-hazard platform integrating landslide prediction, early warning decision support, and emergency response resource coordination for Sri Lanka. As part of the **Landslide Prediction and Forecasting Component** (conducted by Bagya R M S, IT23394124), we require soil classification data to analyze slope stability, soil composition, and geotechnical vulnerability.

We are writing to formally request access to the **Soil Type / Soil Classification Dataset** maintained by the Department of Agriculture.

1. Scope of Request

- **Dataset Requested:** Soil Classification / Soil Type Maps (e.g., Red-Yellow Podzolic, Mountain Regosol, Alluvial, Latosols, and related soil series/associations)
- **Geographical Level:** District, Divisional Secretariat (DS) Division, and Grama Niladhari (GN) Division levels, or mountain-block level

- **Geographic Focus:** Nuwara Eliya District (specifically Ambagamuwa, Kotmale, Walapane, Kelanai river and Kalu river DS Divisions) for high-resolution landslide modeling, alongside island-wide coverage where available

2. Specific Variables Requested

Category	Specific Variables Requested
Soil Classification & Mapping	Soil type boundaries, soil unit classifications (e.g., Red-Yellow Podzolic, Mountain Regosol), soil texture/composition classes, drainage characteristics, and depth profiles where available.
Administrative / Spatial References	Geo-referenced spatial boundary files (Vector polygon/shapefile) mapped across District, DS, and GN Division boundaries.

3. Purpose and Data Use

The requested soil dataset will be used strictly for academic research purposes: specifically as a core feature input in machine learning models to identify clay-trap zones versus colluvium zones for slope stability and landslide susceptibility modeling. All data will be handled in accordance with SLIIT's research ethics and data privacy guidelines, used solely within the scope of this project, and not shared with third parties or used for commercial purposes. We are glad to sign any data-usage agreement or non-disclosure undertaking the Department may require.

4. Format

We would be grateful to receive the data in a machine-readable GIS format—such as **Shapefile (.shp)**, **GeoJSON**, or **Vector Polygon Layers**—or in any digitally accessible format convenient for your team.

We would be most grateful for your kind consideration of this request. Please do not hesitate to contact us or our supervisors should any clarification be required.

Thank you for your time and support.

Yours faithfully,

Dr. Kapila Dissanayaka

Supervisor, SLIIT

Mr. Uchira Wickramarathne

Co-Supervisor, SLIIT

Bagya R M S (IT23394124)

Undergraduate Student, SLIIT